Ambient Polymorphism

Sam Estep

Abstract

…

Introduction

PostgreSQL 11.22 (2023) returns zero for the following query (National Institute of Standards and Technology, n.d.):

NumAcc4.sql

```
SELECT stddev(
  CASE
    WHEN k = 1 THEN 10000000.2
    WHEN k % 2 = 0 THEN 10000000.1
    ELSE 10000000.3
  END :: double precision
)
FROM generate_series(1, 1001) AS k;
```

Why? Up until version 12, Postgres computed standard deviation by first computing $a = \sum x$ and $b = \sum x^2$ …

float.c

```
N += 1.0;
sumX += newval;
CHECKFLOATVAL(sumX, isinf(transvalues[1]) || isinf(newval), true);
sumX2 += newval * newval;
CHECKFLOATVAL(sumX2, isinf(transvalues[2]) || isinf(newval), true);
```

… and then computing $Nb - a^2$:

float.c

```
numerator = N * sumX2 - sumX * sumX;
CHECKFLOATVAL(numerator, isinf(sumX2) || isinf(sumX), true);

/* Watch out for roundoff error producing a negative numerator */
if (numerator <= 0.0)
    PG_RETURN_FLOAT8(0.0);
```

This catastrophic cancellation is

> the notorious desk top calculator algorithm, also known as the one-pass algorithm.
>
> — Simon and Lesage (1988)

> Higher precision will very often estimate a “correct” path well enough. To this end, imagine a debugger that can transform a given subprogram p, whose literal constants and variables x, y, z, … have been declared by the programmer to have precisions thought adequate at the time, into an analogous subprogram P whose corresponding literal constants and variables X, Y, Z, … are declared by the debugger to have greater precisions, preferably about twice as great. Then the debugger can execute both programs p and P simultaneously (actually interleaved) with the same input (or copies of it if the subprogram will change it) and compare their progress to see where one of the variables x, y, z, … first departs excessively from its analog X, Y, Z, … .

> — Kahan (2006)

Bibliography

Kahan, William. 2006. “How Futile Are Mindless Assessments of Roundoff in Floating-Point Computation?.” January 11. https://people.eecs.berkeley.edu/~wkahan/Mindless.pdf.

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